

Practical no. 9 AI
Association Rule Mining

AIM : Implement the Association Rule Mining algorithm (e.g., Apriori) to find frequent itemsets.

Generate association rules from the frequent itemsets and calculate their support and confidence.

Interpret and analyze the discovered association rules.

Code :

```
ai p9 T117.py - C:\Users\Adarsh\OneDrive\Videos\ai p9 T117.py (3.12.0)
File Edit Format Run Options Window Help
import pandas as pd
import matplotlib.pyplot as plt
from itertools import combinations

print("Adarsh Shukla T117")

df = pd.read_csv(r"C:\Users\Adarsh\Downloads\retail_sales_dataset.csv")

print("\nDataset:")
print(df.head())

print("\nColumns:")
print(df.columns.tolist())

transactions = []

for _, row in df.iterrows():
    basket = []

    for col in df.columns:
        name = col.lower()

        if "transaction" in name or "customer id" in name or name == "date":
            continue

        value = row[col]

        if pd.isna(value):
            continue

        if col.lower() == "age":
            age = int(value)
```

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    basket.append("Age=Senior")

    elif "price" in name or "amount" in name:
        continue

    else:
        basket.append(f"{col}={value}")

    transactions.append(set(basket))

min_support = 0.05
min_confidence = 0.5

total_transactions = len(transactions)

all_items = sorted(set().union(*transactions))

frequent_itemsets = {}

for item in all_items:
    support = sum(item in t for t in transactions) / total_transactions

    if support >= min_support:
        frequent_itemsets[frozenset([item])] = support

current_level = list(frequent_itemsets.keys())
k = 2

while current_level:
    previous_items = sorted(set().union(*current_level))

    candidates = list(combinations(previous_items, k))

    next_level = []

    for candidate in candidates:
        candidate_set = frozenset(candidate)

        support = sum(
            candidate_set.issubset(transaction)
            for transaction in transactions
        ) / total_transactions

        if support >= min_support:
            frequent_itemsets[candidate_set] = support
            next_level.append(candidate_set)

    current_level = next_level
    k += 1

itemset_results = []

for itemset, support in frequent_itemsets.items():
    if len(itemset) >= 2:
        itemset_results.append({
            "Itemset": " , ".join(sorted(itemset)),
            "Support": support
        })
```

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    "Itemset": " , ".join(sorted(itemset)),
    "Support": support
})

itemset_df = pd.DataFrame(itemset_results)

if not itemset_df.empty:
    itemset_df = itemset_df.sort_values(
        "Support",
        ascending=False
    ).reset_index(drop=True)

print("\nFrequent Itemsets:")
if itemset_df.empty:
    print("No frequent itemsets found.")
else:
    print(itemset_df.head(20).to_string(index=False))

rules = []

for itemset, itemset_support in frequent_itemsets.items():
    if len(itemset) < 2:
        continue

    for size in range(1, len(itemset)):
        for antecedent_tuple in combinations(itemset, size):
            antecedent = frozenset(antecedent_tuple)
            consequent = itemset - antecedent
```

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plt.barh(
    top_items["Itemset"],
    top_items["Support"]
)

plt.xlabel("Support")
plt.ylabel("Frequent Itemsets")
plt.title("Top Frequent Itemsets")

plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()

if not rules_df.empty:

    plt.figure(figsize=(8, 6))

    plt.scatter(
        rules_df["Support"],
        rules_df["Confidence"],
        s=rules_df["Lift"] * 80,
        alpha=0.7
    )

    plt.xlabel("Support")
    plt.ylabel("Confidence")
    plt.title("Association Rules: Support vs Confidence")

    plt.grid(True)
    plt.tight_layout()
    plt.show()

print("\nAssociation Rules:")

if rules_df.empty:
    print("No association rules found.")
else:
    print(rules_df.head(20).to_string(index=False))

    print("\nBest Rule:")
    print(
        rules_df.iloc[0]["Antecedent"],
        " -> ",
        rules_df.iloc[0]["Consequent"]
    )

    print(
        "Support:",
        round(rules_df.iloc[0]["Support"], 3)
    )

    print(
        "Confidence:",
        round(rules_df.iloc[0]["Confidence"], 3)
    )

    print(
        "Lift:",
        round(rules_df.iloc[0]["Lift"], 3)
    )

if not itemset_df.empty:
    top_items = itemset_df.head(10)
```

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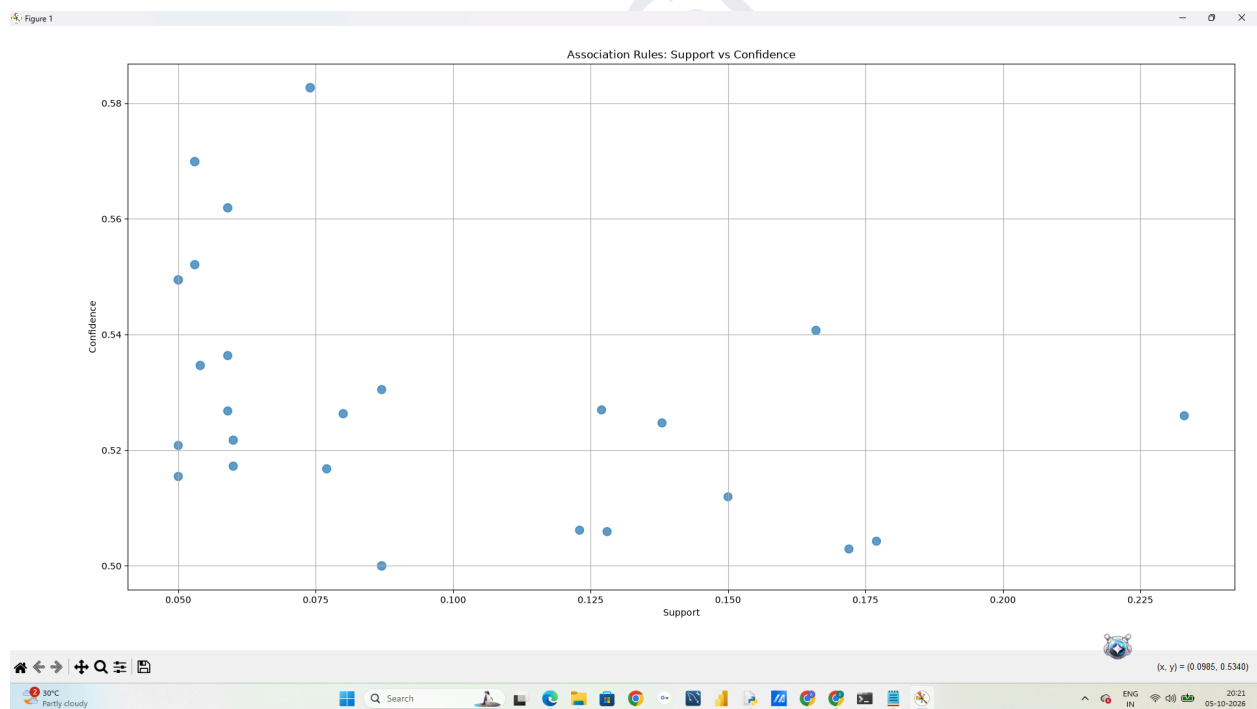
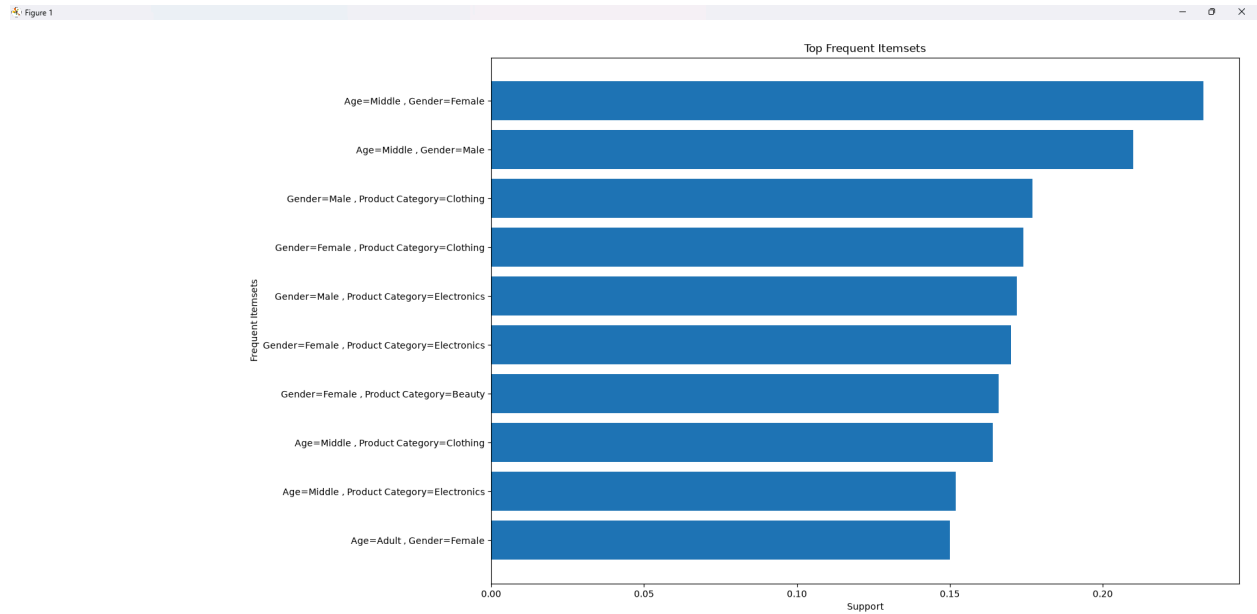
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```

Python 3.12.6 (tags/v3.12.6:a4a2d2b, Sep  6 2024, 20:11:23) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Adarsh/OneDrive/Videos/ai p9 T117.py
Adarsh Shukla T117

Dataset:
  Transaction ID    Date    ... Price per Unit Total Amount
0                1  2023-11-24    ...              50          150
1                2  2023-02-27    ...             500         1000
2                3  2023-01-13    ...              30           30
3                4  2023-05-21    ...             500          500
4                5  2023-05-06    ...              50          100

[5 rows x 9 columns]

Columns:
['Transaction ID', 'Date', 'Customer ID', 'Gender', 'Age', 'Product Category', 'Quantity', 'Price per Unit', 'Total Amount']

Frequent Itemsets:
      Itemset    Support
      Age=Middle , Gender=Female    0.233
      Age=Middle , Gender=Male    0.210
      Gender=Male , Product Category=Clothing    0.177
      Gender=Female , Product Category=Clothing    0.174
      Gender=Male , Product Category=Electronics    0.172
      Gender=Female , Product Category=Electronics    0.170
      Gender=Female , Product Category=Beauty    0.166
      Age=Middle , Product Category=Clothing    0.164
      Age=Middle , Product Category=Electronics    0.152
      Age=Adult , Gender=Female    0.150

Columns:
['Transaction ID', 'Date', 'Customer ID', 'Gender', 'Age', 'Product Category', 'Quantity', 'Price per Unit', 'Total Amount']

Frequent Itemsets:
      Itemset    Support
      Age=Middle , Gender=Female    0.233
      Age=Middle , Gender=Male    0.210
      Gender=Male , Product Category=Clothing    0.177
      Gender=Female , Product Category=Clothing    0.174
      Gender=Male , Product Category=Electronics    0.172
      Gender=Female , Product Category=Electronics    0.170
      Gender=Female , Product Category=Beauty    0.166
      Age=Middle , Product Category=Clothing    0.164
      Age=Middle , Product Category=Electronics    0.152
      Age=Adult , Gender=Female    0.150
      Age=Adult , Gender=Male    0.143
      Gender=Male , Product Category=Beauty    0.141
      Gender=Female , Quantity=4    0.138
      Gender=Male , Quantity=1    0.128
      Age=Middle , Product Category=Beauty    0.127
      Gender=Female , Quantity=3    0.127
      Gender=Female , Quantity=1    0.125
      Gender=Male , Quantity=4    0.125
      Gender=Male , Quantity=2    0.123
      Gender=Female , Quantity=2    0.120

Association Rules:
      Antecedent    Consequent    Support    Confidence    Lift
      Age=Middle , Product Category=Beauty    Gender=Female    0.074    0.582677    1.142504
      Product Category=Electronics , Quantity=4    Gender=Female    0.053    0.569892    1.117436
  
```

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```
"IDLE Shell 3.12.6"
File Edit Shell Debug Options Window Help

Gender=Male , Quantity=2    0.123
Gender=Female , Quantity=2   0.120

Association Rules:

Antecedent      Consequent      Support      Confidence      Lift
Age=Middle , Product Category=Beauty Gender=Female 0.074 0.582677 1.142504
Product Category=Electronics , Quantity=4 Gender=Female 0.053 0.569892 1.117436
Age=Middle , Quantity=3 Gender=Female 0.059 0.561905 1.101774
Age=Adult , Product Category=Electronics Gender=Female 0.053 0.552083 1.082516
Product Category=Electronics , Quantity=1 Gender=Male 0.050 0.549451 1.121328
Product Category=Beauty Gender=Female 0.166 0.540717 1.060229
Age=Middle , Quantity=2 Gender=Female 0.059 0.536364 1.051693
Age=Adult , Product Category=Clothing Gender=Male 0.054 0.534653 1.091130
Age=Middle , Product Category=Clothing Gender=Female 0.087 0.530488 1.040172
Quantity=3 Gender=Female 0.127 0.526971 1.033276
Age=Middle , Quantity=4 Gender=Female 0.059 0.526786 1.032913
Age=Middle , Product Category=Electronics Gender=Male 0.080 0.526316 1.074114
Age=Middle Gender=Female 0.233 0.525959 1.031293
Quantity=4 Gender=Female 0.138 0.524715 1.028853
Age=Senior Gender=Male 0.060 0.521739 1.064774
Age=Adult , Product Category=Beauty Gender=Female 0.050 0.520833 1.021242
Age=Middle , Quantity=1 Gender=Male 0.060 0.517241 1.055595
Age=Young Gender=Male 0.077 0.516779 1.054650
Product Category=Clothing , Quantity=4 Gender=Male 0.050 0.515464 1.051967
Age=Adult Gender=Female 0.150 0.511945 1.003814

Best Rule:
Age=Middle , Product Category=Beauty -> Gender=Female
Support: 0.074
Confidence: 0.583
Lift: 1.143
```

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